A Joint Multiscale Graph Attention and Classify-Driven Autoencoder Framework for Hyperspectral Unmixing

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***Abstract*— Deep learning has recently gained popularity in hyperspectral unmixing (HU) and typical methods involve convolutional neural network-based (CNN-based) and autoencoding-based methods. However, most existing methods are usually confined to capturing local features in hyperspectral images (HSIs) while neglecting long-range dependency information on spatial position in HSIs, where long-range dependency on spatial positions means the correlations between spatial pixels or regions. Graph neural networks (GNNs) have recently shown great potential in various fields, which model complex spatial relationships and interactions in data. Therefore, this article develops a joint multiscale graph attention and classify-driven autoencoder (MSGA-CD) framework for HU. Its core is to construct a multiscale graph attention abundance (MSGAA) module, a local–global abundance fusion (LGAF) module, and an abundance-classify-driven endmember decoder (ACDE) module. Concretely, MSGAA incorporates a multiscale strategy into the graph attention network (GAT) to extract diverse long-range dependency information on spatial locations in HSIs from different levels and obtain global abundances. Afterward, LGAF integrates local abundance obtained by CNN and global abundance by MSGAA, achieving a more precise abundance representation. Moreover, ACDE clusters all HSI pixel features into various endmember categories using abundance fractions and takes them as priors to drive endmember learning, effectively improving the accuracy of endmember extraction. Finally, the abundance and endmember matrices are trained simultaneously by constraining their dependent relationship through a joint loss. Experiments reveal that MSGA-CD outperforms state-of-the-art methods, offering a promising method for HU.**

***Index Terms*— Deep learning, feature extraction, graph neural networks (GNNs), hyperspectral unmixing (HU), multiscale.**

I. Introduction

HYPERSPECTRAL images (HSIs), containing hundreds or even thousands of continuous bands, have complex spectral details and highlight surface features. These HSIs have been widely used in a variety of applications, such as target detection [1], [2], environmental monitoring [3], [4], and urban planning [5], [6]. However, due to the limited spatial resolution of hyperspectral sensors, some mixed pixels with multiple materials occur in the obtained HSI. To acquire fine land cover information from complex mixed pixels, hyperspectral unmixing (HU) is required. It seeks to separate these mixed pixels by identifying pure materials (called endmembers) in mixed pixels and calculating their abundances (the proportion of each pure material within each mixed pixel).

Traditional HU methods have been extensively studied. Classic approaches include geometric-based methods [7], [8], statistical-based methods [9], nonnegative matrix factorization-based methods [10], [11], [12], sparse representation-based methods [13], [14], graph Laplacian weighted robust sparse unmixing [15], as well as some nonlinear spectral unmixing methods [16], [17]. However, these methods typically built optimization models and algorithms to process HSI mixed pixels and often relied on specific assumptions and the past understanding of experts, which may affect their effectiveness.

Deep learning has advanced rapidly due to its strong feature extraction capabilities, significantly impacting many tasks involving HSIs [18], [19], [20]. In HU, it has gained significant interest as well [21], [22]. Deep learning-based HU methods are divided into two categories: supervised and unsupervised HU. The former requires known endmembers to estimate abundance fractions. For example, Khajehrayeni and Ghassemian [23] estimated the abundance using a deep convolutional autoencoder (AE) when the spectral information of endmembers was known. Shen et al. [24] used all of the data to ensure little loss of prior knowledge. In contrast, unsupervised HU, also referred to as blind unmixing [25], has gained increased attention in many real-world scenarios involving unknown endmembers. This article will emphasize unsupervised deep learning-based HU methods.

Unsupervised HU typically consists of two steps: encoding and decoding. The original HSI is encoded and decoded to derive the abundance and endmembers, respectively. One popular unsupervised HU strategy is based on the linear mixing model (LMM) [26]. It assumes that the spectral signature of mixed pixels is a linear combination of endmember signatures, making it a wide choice in existing unmixing methods due to its simplicity and ease of interpretation. For instance, Guo et al. [27] incorporated AEs into HU by utilizing nonnegative sparse AEs and denoising AEs to improve abundance estimation. Similarly, Su et al. [28] applied variational AEs to effectively extract endmembers and abundances after employing stacked AEs to learn spectral features.

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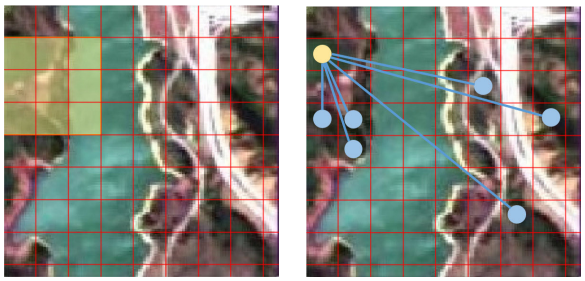
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Min and Li [29] proposed a probability metric-based AE approach. Jin et al. [30] integrated superpixel segmentation and pseudo-pure pixel learning within a two-stream AE network. Min et al. [31] enhanced the distribution similarity and feature differences by adding Wasserstein distance and feature-matching regularization to the loss. However, most AE-based methods predominantly utilize spectral information while neglecting the spatial information inherent in HSIs. To this end, Palsson et al. [32] introduced convolutional neural networks (CNNs) into AEs to better capture spatial and spectral information of HSIs. Gao et al. [33] developed a cycle-consistent unmixing network (CyCU-Net) based on self-perception to prioritize material-related information. Yu et al. [34] proposed a multistage convolutional AE network (MSNet) to emphasize material distribution and contextual information. Zhao et al. [35] enhanced unmixing performance by integrating deep AE features with prior information provided by a denoiser. Ghosh et al. [36] incorporated Transformer architectures into convolutional AEs to effectively capture contextual information. Lastly, Borsoi et al. [37] modeled the temporal evolution of endmembers and abundances in HSI sequences using variational recurrent neural networks. Jin et al. [38] applied adversarial networks to AEs to mitigate the impacts of noise and initialization issues. Another unsupervised HU strategy is based on the nonlinear mixing model (NLMM) [39]. It considers complex interactions between solar radiation and various materials, providing a more comprehensive description of scattering phenomena. Wang et al. [40] employed a nonlinear AE structure to capture scattering effects in HSIs, enhancing the understanding of intricate light–material interactions. Rasti et al. [41] integrated NLMM with a fully convolutional encoder–decoder network, utilizing loss to estimate endmembers and abundances in intimate mixtures. Li et al. [42] utilized a coarse-to-fine scheme to jointly estimate endmembers and abundances, incorporating weighted sparse and graph regularizers to model prior knowledge. Recently, Su et al. [43] developed a deep AE-based augmented network (DAAN) that integrated NLMM with AE to consider material interactions and spatial scattering effects. A more comprehensive review of HSI unmixing methods can be found in [44]. This article will develop an unsupervised unmixing method based on LMM.

Although these works have achieved certain effectiveness, they still have shortcomings.

1) Most convolutional neural network-based (CNN-based) and AE-based methods are limited to capturing local spatial information for HU, as shown in Fig. 1(a). They easily ignore long-range dependency information on spatial positions in HSI, which refers to the relationships between spatial pixels or areas. In contrast, graph neural networks (GNNs) can interact with those areas without directly adjacent in space, as shown in Fig. 1(b). As such, GNN can extract long-range dependency information on spatial positions in HSI for HU. To date, there have been few GNN works on HU. Therefore, this article emphasizes integrating GNN into HU. Although Jin et al. [39] incorporated a graph attention network (GAT) into AE, it extracted features from HSIs using a single-scale superpixel segmentation strategy. This may not effectively segment HSIs into areas with similar spectral features, ignoring different levels of information and lacking feature diversity. The overall unmixing performance may potentially be affected.
2) Previous methods treat network weights as endmembers through a simple fully connected layer or 1×1 convolutional layer in the decoder phase and rely on relatively rough weight learning and linear constraints of pixels to obtain endmembers. However, these methods overlook the inherent complexity of ground objects in HSIs, affecting the effectiveness of extracting reliable endmember information.



(a) (b)

Fig. 1. Comparison between CNN and GNN. (a) CNN conducts a regular convolution operation. It usually only captures local information due to the limited receptive field but ignores long-range information on spatial positions in HSIs. (b) GNN performs a graph convolution operation, with each area represented by a node and the blue line representing an edge. The formed graph structure can interact with long-range dependency information on spatial positions in HSIs.

To this end, this article proposes a joint multiscale graph attention and classify-driven autoencoder (MSGA-CD) framework for unsupervised HU. Its core is to devise a multiscale graph attention abundance (MSGAA) module, a local–global abundance fusion (LGAF) module, and an abundance-classify-driven endmember decoder (ACDE) module. Specifically, to obtain different scales of information and produce diverse long-range dependency features on spatial positions in HSIs, an MSGAA module is constructed. It incorporates multiple segmentation scales of HSIs into GAT, enriching the global abundance. Here, the resulting abundance information, obtained by MSGAA to reflect diverse long-range dependencies in global spatial positions within the HSI, is referred to as global abundance. Meanwhile, a convolution module is applied to the original HSIs, extracting local spatial information. The abundance information derived from these local spatial features is termed local abundance. Afterward, to effectively integrate the global and local abundance information, an LGAF module is designed based on a gating mechanism, providing a more refined and comprehensive abundance representation. Moreover, an ACDE module is presented. It clusters all mixing pixels of the original HSI using the maximum element strategy of abundance fraction vectors, with the number of clusters equal to the number of endmembers P. This strategy treats pixel features in each cluster as a group (a total of P groups). Then, it guides the extraction process of each

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endmember type by constructing a learnable network, enabling effective learning of all endmembers.

Compared with the previous GNN work [39] for HU, the proposed MSGA-CD framework differs in several aspects. First, this article introduces a multiscale GAT strategy to acquire diverse global abundance information directly, whereas [39] used a single-scale GAT to extract global spatial features of HSI rather than global abundance information. Therefore, their roles are completely inconsistent. Second, this article devises an abundance-classify-driven endmember decoder, which clusters each original HSI mixing pixel feature using the maximum element strategy of abundance fraction vectors to drive the generation of endmembers. In contrast, Jin and Yang [39] used a polynomial posterior nonlinear mixing module based on the bilinear mixing model to obtain endmembers. Therefore, their ideas are distinct. Overall, the main contributions of this article are summarized as follows.

1) A novel MSGA-CD framework is presented to produce comprehensive abundances and accurate endmembers, effectively improving the performance of HU.
2) An MSGAA module is designed. It integrates a multiscale GAT strategy to promote feature diversity and capture global abundance information at different scales.
3) An LGAF module is constructed. It uses two branches to interact local and global abundance information with each other, generating more comprehensive abundance fractions.
4) A new ACDE module is devised, which uses the maximum element strategy of abundance fraction vectors to cluster HSI mixing pixels and drive endmember learning, improving the accuracy of endmember learning.

The rest of this article is structured as follows. Section II gives preliminaries. Section III presents a detailed description of MSGA-CD. Section IV conducts experiments and analyses. Finally, Section V summarizes the article.

II. Preliminaries

A. Linear Mixing Model

This article focuses on the LMMs, which refer to each pixel in an HSI being decomposed into the product of endmembers and abundances linearly. Specifically, let $\boldsymbol{Y}_{\text{ori}} \in \mathbb{R}^{B\times H\times W}$ is the original HSI, where B, H, and W are the number of spectral bands, the height, and the width of $\boldsymbol{Y}_{\text{ori}}$, respectively, then LMM is expressed as

$$\boldsymbol{Y} = \boldsymbol{M}\boldsymbol{S} + \varepsilon \tag{1}$$

where $\boldsymbol{Y} = (\boldsymbol{Y}_1, \boldsymbol{Y}_2, \ldots, \boldsymbol{Y}_N) \in \mathbb{R}^{B\times N}$ is the observed HSI with $N = HW$ pixels along the lexicographical order on $\boldsymbol{Y}_{\text{ori}}$, $\boldsymbol{M} = (\boldsymbol{M}_1, \boldsymbol{M}_2, \ldots, \boldsymbol{M}_P) \in \mathbb{R}^{B\times P}$ is the endmember matrix representing the spectral information of P endmembers, $\boldsymbol{S} = (S_{ij}) \in \mathbb{R}^{P\times N}$ is the corresponding abundance matrix, and $\varepsilon \in \mathbb{R}^{B\times N}$ is noise.

Since endmembers and abundances are the spectral information and related proportion of each material in mixed pixels, respectively, (1) needs to satisfy the endmember nonnegativity constraint, the abundance nonnegativity constraint (ANC), and the abundance sum-to-one constraint (ASC), i.e.,

$$\begin{cases} \boldsymbol{M}_i & \succeq 0 \\ S_{ij} & \geq 0, \quad \sum_{i=1}^{P} S_{ij} = 1 \end{cases} \tag{2}$$

where "$\succeq$" represents that each component of a vector should be more than 0.

B. AE-Based Unmixing Framework

The AE network is a classic framework for HU tasks [45], [46]. Its basic structure consists of an encoder and a decoder.

1) *Encoder:* It maps the input hyperspectral data $\boldsymbol{Y}$ into a low-dimensional latent space, which is treated as an abundance matrix $\boldsymbol{S}$ physically, defined as

$$\boldsymbol{S} = f_e(\boldsymbol{Y}) \tag{3}$$

where $f_e(\cdot)$ denotes an encoder operation.

2) *Decoder:* For HU tasks, it maps this low-dimensional representation back to the original data space to produce reconstructed HSI data, given by

$$\hat{\boldsymbol{Y}} = f_d(\boldsymbol{W}, \boldsymbol{S}) \tag{4}$$

where $f_d(\cdot)$ denotes an encoder operation, and $\hat{\boldsymbol{Y}}$ is the reconstructed hyperspectral data. $\boldsymbol{W}$ is the weight of the decoder layer, which usually uses a fully connected [27] or 1×1 convolutional layer [47] in LMM. At this time, $\boldsymbol{W}$ is modeled as endmembers $\boldsymbol{M}$.

C. Graph Attention Network

GNNs have been widely used in various fields, such as HSI classification [48], [49], medical data analysis [50], and recommendation [51], [52]. Here, we review a classic GAT model [53]. Given a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, with $\mathcal{V}$ and $\mathcal{E}$ representing the node and edge sets, respectively. The input feature of the graph is denoted as $\boldsymbol{X} = \{\boldsymbol{x}_i\}_{i=1}^{\hat{N}} \in \mathbb{R}^{\hat{B}\times\hat{N}}$, where $\boldsymbol{x}_i$ represents the ith node with the feature dimension $\hat{B}$, and $\hat{N}$ is the number of nodes. The relationship between nodes is represented by an adjacency matrix $\boldsymbol{A} = (a_{ij}) \in \mathbb{R}^{\hat{N}\times\hat{N}}$. GAT introduces an attention mechanism to assign weights to each neighbor and better capture the complex relationships between nodes. Mathematically, the importance e_{ij} between the ith node and the jth node is formulated as

$$e_{ij} = \text{LeakyReLU}\left(\boldsymbol{a}^T\left[\boldsymbol{\Theta}\boldsymbol{x}_i || \boldsymbol{\Theta}\boldsymbol{x}_j\right]\right) \tag{5}$$

where $\boldsymbol{a}$ is a shared learnable vector, $\boldsymbol{\Theta}$ denotes a learnable matrix, and $||$ is a concatenation operation. After that, the softmax function is used to normalize it, described as

$$\alpha_{ij} \triangleq \text{softmax}(e_{ij}) = \frac{\exp(e_{ij})}{\sum_{k\in\mathcal{N}_i}\exp(e_{ik})} \tag{6}$$

where $\mathcal{N}_i = \{j \mid a_{ij} = 1\}$ is the neighborhood of the ith node in the graph. Afterward, a nonlinear activation function σ (such as ReLU) is applied to obtain output features of the ith node through a GAT layer, i.e.,

$$\boldsymbol{x}_i' = \sigma\left(\sum_{j\in\mathcal{N}_i}\alpha_{ij}\boldsymbol{\Theta}\boldsymbol{x}_j\right). \tag{7}$$

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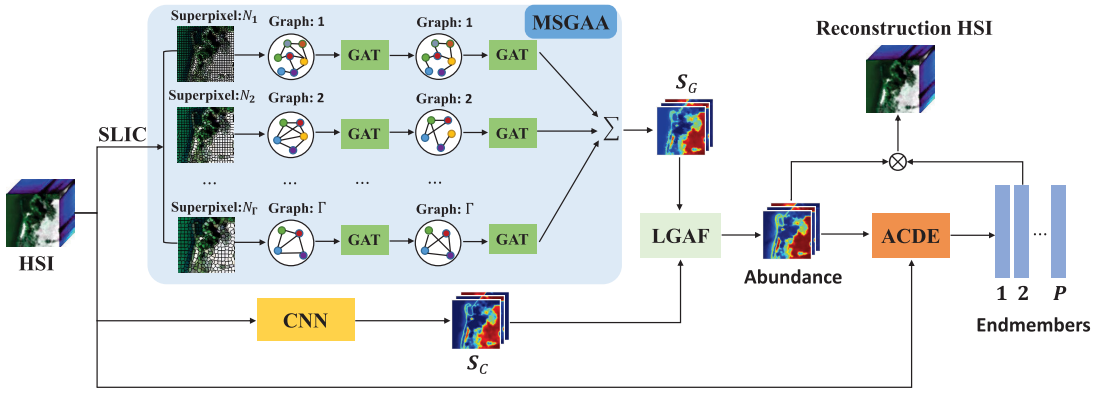


Fig. 2. Overall structure of MSGA-CD. SLIC: Simple linear iterative clustering; MSGAA: Multiscale graph attention abundance; LGAF: Local–global abundance fusion; ACDE: Abundance-classify-driven endmember decoder.

III. Proposed Method

A. Overview of MSGA-CD

Fig. 2 shows the overall process of MSGA-CD. It mainly includes an MSGAA module, an LGAF module, and an ACDE module. First, MSGAA is constructed by integrating different segmentation scales of HSIs into GAT, which capture the diverse long-range dependency on spatial positions in HSI and enrich global abundance information $\boldsymbol{S}_G$, defined as

$$\boldsymbol{S}_G = f(\boldsymbol{Y}) \tag{8}$$

where $f(\cdot)$ is the proposed MSGAA module. Further, CNN is used to learn local abundance information $\boldsymbol{S}_C$. Afterward, to acquire accurate abundance fractions, we fuse both the local and global abundance information through the LGAF module $g(\cdot)$ based on the gated mechanism, i.e.,

$$\boldsymbol{S} = g(\boldsymbol{S}_G, \boldsymbol{S}_C) \tag{9}$$

where $\boldsymbol{S}$ is the final abundance matrix after LGAF. Lastly, to effectively improve the accuracy of endmember extraction, an ACDE module is designed to cluster all mixing pixels of HSIs using the maximum element strategy of abundance fraction vectors and regulate the endmember extraction, i.e.,

$$\boldsymbol{M} = h(\boldsymbol{Y}, \boldsymbol{S}) \tag{10}$$

where $h(\cdot)$ is the ACDE module and $\boldsymbol{M}$ is the endmember matrix that is consistent with (1). Next, we will present these three modules in detail.

B. MSGAA Module

Previous HU methods focus more on local information and ignore the long-range dependency information on spatial positions HSIs. GNN effectively collects this information using pixel correlation in global spatial positions of HSI. But using a GNN with a single-scale superpixel segmentation strategy is difficult to effectively segment HSIs into regions with similar spectral features, neglecting various levels of information and lacking feature diversity. To this end, this article proposes an MSGAA module, which uses a multiscale learning strategy, to fully mine the diversity of features on HSIs and provide more accurate global abundance. However, a critical issue is to build a graph structure (nodes and edges) for HSI when using GNNs. Although using each pixel as a node is simple, the large number of pixels in HSIs significantly increases the complexity of subsequent operations, limiting the capabilities of GNNs. In contrast, superpixels are used as nodes to reduce graph size while preserving the local features of HSI. Accordingly, this article constructs superpixels as nodes.

Specifically, we adopt Γ scales to form multiscale graph structures, with each scale segmenting the HSI by a simple linear iterative clustering (SLIC) [54] and generating N_τ superpixels as all nodes of the τth graph ($\tau = 1, 2, \ldots, \Gamma$). The spectral signature of each superpixel is calculated by averaging spectral information of all pixels within the superpixel and taking it as a feature of each node, denoted as $\boldsymbol{X}_\tau = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_{N_\tau}] \in \mathbb{R}^{B\times N_\tau}$. For edges in the τth graph, we first consider superpixels directly adjacent in spatial location as neighbors for a given center node i. Further, we also evaluate the correlations between the i-th superpixel (node) feature $\boldsymbol{x}_i$ ($i = 1, 2, \ldots, N_\tau$) and the rest superpixels by calculating the cosine similarity Ω_{ij} as

$$\Omega_{ij} = \frac{\boldsymbol{x}_i^T \boldsymbol{x}_j}{\|\boldsymbol{x}_i\|_2 \|\boldsymbol{x}_j\|_2}, \quad j \neq i \tag{11}$$

where $\|\cdot\|$ is the Euclidean norm. For the ith node, we choose the k highest similarity nodes as neighbors. By combining these neighbors with the spatially adjacent neighbors, we construct an adjacency matrix $\boldsymbol{A}_\tau = (a_{ij}^\tau) \in \mathbb{R}^{N_\tau \times N_\tau}$ for the τth graph, where $a_{ij}^\tau = 1$ if j is the neighbor of i, otherwise $a_{ij}^\tau = 0$. After that, we use two-layer GAT to update node features using $\boldsymbol{X}_\tau$ and $\boldsymbol{A}_\tau$ as inputs, i.e.,

$$\boldsymbol{X}'_\tau = \mathrm{GAT}(\mathrm{GAT}(\boldsymbol{X}_\tau, \boldsymbol{A}_\tau)), \ \tau = 1, 2, \ldots, \Gamma \tag{12}$$

where $\mathrm{GAT}(\cdot, \cdot)$ is the GAT layer by updating (7) for each node in Section II-C, with the activation function σ of the first and the second GAT layers be the ReLU and Softmax functions, respectively.

Therefore, $\boldsymbol{X}'_\tau \in \mathbb{R}^{P\times N_\tau}$ satisfies the ANC and ASC conditions. It is the abundance matrix of all nodes. Because each superpixel block contains several pixels, we need to create an association matrix $\boldsymbol{Q} = (Q_{ni}) \in \mathbb{R}^{N\times N_\tau}$ that describes the relationship between each pixel and the superpixel block to

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which it belongs. For each pixel, we have

$$Q_{ni} = \begin{cases} 1, & \text{if } \boldsymbol{Y}_n \in \mathcal{M}_i \\ 0, & \text{otherwise} \end{cases} \tag{13}$$

where $\boldsymbol{Y}_n \in \mathbb{R}^{B\times 1}$ is the nth ($n = 1, 2, \ldots, N$) pixel on the HSI, $\mathcal{M}_i$ denotes the ith ($i = 1, 2, \ldots, N_\tau$) superpixel set. Then, the obtained superpixel abundance matrix $\boldsymbol{X}'_\tau$ is converted to a pixel abundance matrix by

$$\boldsymbol{S}_\tau = \boldsymbol{X}'_\tau \boldsymbol{Q}^T \tag{14}$$

where $\boldsymbol{S}_\tau$ is the abundance matrix of all mixed pixels for the τth scale. Since each pixel only belongs to one superpixel, $\boldsymbol{S}_\tau$ still meets ANC and ASC constraints. For Γ scales, we use learnable weights $\{w_\tau\}_{\tau=1}^{\Gamma}$ to adaptively calculate the weighted sum of Γ abundance matrices as

$$\boldsymbol{S}_G = \sum_{\tau=1}^{\Gamma} w_\tau \boldsymbol{S}_\tau \tag{15}$$

where $\sum_{\tau=1}^{\Gamma} w_\tau = 1$. Therefore, $\boldsymbol{S}_G \in \mathbb{R}^{P\times N}$ is the global abundance matrix, satisfying the ANC and ASC constraints.

Overall, MSGAA effectively captures long-range dependency information on spatial positions in HSIs and directly extracts global abundance information through multiscale GAT instead of feature information. It better learns different scale features in HSI and reflects the diversity of features compared with a single scale.

C. LGAF Module

Though MSGAA acquires global abundance, local features of HSIs are also crucial. To this end, we use a CNN (using three convolutional layers with each following an activation function) to extract local abundance $\boldsymbol{S}_C \in \mathbb{R}^{P\times N}$ from original HSIs. The process is described as

$$\begin{cases} \hat{\boldsymbol{S}}_C = \text{Conv}(\phi(\text{Conv}(\phi(\text{Conv}(\boldsymbol{Y}_{\text{ori}}))))) \\ \boldsymbol{S}_C = \text{Softmax}\big(\text{Reshape}\big(\hat{\boldsymbol{S}}_C\big)\big) \end{cases} \tag{16}$$

where Conv is a 2D convolution layer, $\phi(\cdot)$ is a LeakyReLU activation function, Reshape$(\cdot)$ is a transformation along the lexicographical order on $\hat{\boldsymbol{S}}_C$, and Softmax is used to ensure the ANC and ASC constrains.

Afterward, to obtain a comprehensive abundance representation, we design an LGAF module by interacting the global and local abundance information with each other, as shown in Fig. 3. This interaction aims to leverage the complementary advantages of local spatial features captured by CNN and long-range dependency on spatial positions extracted by MSGAA, delivering a more accurate representation to enhance abundance estimation. It involves two branches. First, the upper branch integrates global abundance information into local abundance information, which contributes to improving the accuracy and reliability of local information, described as

$$\begin{cases} \boldsymbol{F}_G = \varphi(\text{Linear}(\text{concat}(\boldsymbol{S}_G, \boldsymbol{S}_C))) \\ \boldsymbol{S}'_C = \alpha(\boldsymbol{F}_G \odot \boldsymbol{S}_G) + \boldsymbol{S}_C \end{cases} \tag{17}$$

where $\varphi(\cdot)$ is a sigmoid function to form an attention matrix in [0, 1], Linear$(\cdot)$ is a fully connected layer, concat$(\cdot)$ is a concatenation operation, α is a learnable parameter, and $\odot$ is element-wise multiplication. Similarly, the lower branch incorporates local abundance information into global abundance information to enhance the accuracy and reliability of global information, defined as

$$\begin{cases} \boldsymbol{F}_C = \varphi(\text{Linear}(\text{concat}(\boldsymbol{S}_C, \boldsymbol{S}_G))) \\ \boldsymbol{S}'_G = \beta(\boldsymbol{F}_C \odot \boldsymbol{S}_C) + \boldsymbol{S}_G \end{cases} \tag{18}$$

where β is a learnable parameter. Finally, we fuse global information $\boldsymbol{S}'_G$ and local information $\boldsymbol{S}'_C$ to obtain the final abundance matrix, i.e.,

$$\boldsymbol{S} = \text{Softmax}\big(\boldsymbol{S}'_G + \boldsymbol{S}'_C\big). \tag{19}$$

Obviously, it still satisfies ANC and ASC constraints.

Overall, the LGAF module effectively integrates global and local abundance information, enhancing abundance information and achieving a more refined and accurate abundance representation.

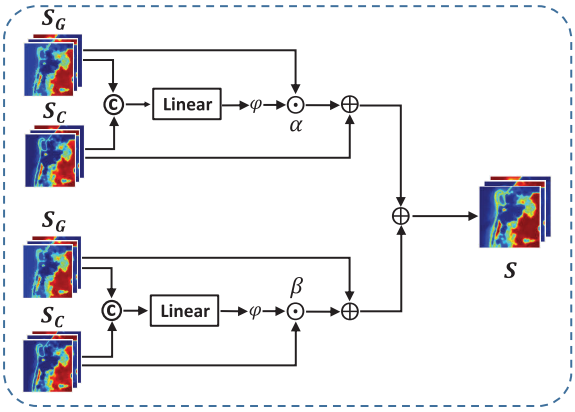


Fig. 3. Structure of LGAF. The upper branch integrates the global abundance $\boldsymbol{S}_G$ into the local abundance $\boldsymbol{S}_C$, while the lower branch incorporates the local abundance $\boldsymbol{S}_C$ into the global abundance $\boldsymbol{S}_G$. Then the obtained information is fused to capture a comprehensive abundance representation $\boldsymbol{S}$.

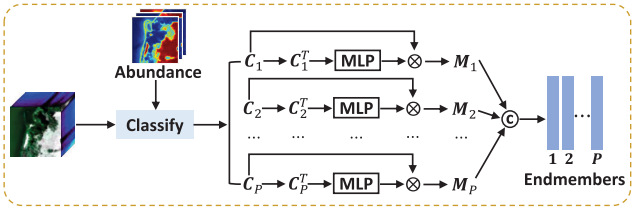


Fig. 4. Structure of ACDE. It first clusters all original mixing pixels into P categories using the maximum element of their abundance fraction vectors to assign class indices. Pixels in each category are then fused to learn a pure endmember feature using MLP. The final endmember matrix is formed by combining these features from P categories.

D. ACDE Decoder Module

In Section II, we mention that most current decoders use the weights of a fully connected layer or 1×1 convolution as endmembers, which rely on simple weight learning and linear constraints of pixels. This strategy ignores the inherent complexity of ground objects in HSIs easily and struggles to extract their endmember information effectively. Therefore, this article constructs an ACDE module, as shown in Fig. 4. Since pixels in the original HSI may contain only

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one substance or have a dominant proportion of a specific substance, we use the spectral information of these pixels as prior knowledge to drive the decoder in more accurately learning the spectral characteristics of endmembers, thereby enhancing the accuracy and reliability of the unmixing results.

First, all mixing pixels are clustered into P categories (i.e., the number of endmembers). In particular, for the ith mixing pixel in the original observed HSI $\boldsymbol{Y}$, we define its class index p using the maximum element of the corresponding abundance fraction vector in the current iteration and then classify the ith mixing pixel feature into $\boldsymbol{C}_p$. This means the ith mixing pixel contains a large proportion of the pth substance. In this way, each mixing pixel feature can be classified into a category set. Then, we obtain all category sets $\{\boldsymbol{C}_p\}_{p=1}^{P}$ (P is the number of endmembers). In other words, all original mixing pixel features $\boldsymbol{Y}$ are transformed into the cluster pixel spectral features, represented as

$$\boldsymbol{Y}_C = \{\boldsymbol{C}_1, \boldsymbol{C}_2, \ldots, \boldsymbol{C}_P\}. \tag{20}$$

Second, each pure endmember is learned from these pixels in the same category $\boldsymbol{C}_p (p = 1, 2, \ldots, P)$, which are fed into a multilayer perceptron (MLP) to generate a shared spectral feature $\boldsymbol{M}_p \in \mathbb{R}^{B\times 1}$ as a pure endmember, represented as

$$\boldsymbol{M}_p = \boldsymbol{C}_p \cdot \mathrm{Softmax}\big(\mathrm{MLP}\big(\boldsymbol{C}_p^{\top}\big)\big), \quad p = 1, 2, \ldots, P. \tag{21}$$

Accordingly, the final endmember matrix is expressed as $\boldsymbol{M} = [\boldsymbol{M}_1, \boldsymbol{M}_2, \ldots, \boldsymbol{M}_P]$.

To sum up, ACDE uses abundance fractions to cluster the pixels of the original HSI. This endmember extraction strategy adequately utilizes the spectral information of the original pixels as priors to guide the endmember learning, promoting the acquisition of the pure spectral feature for each endmember.

E. Loss

This article uses the combination of the spectral angle distance (SAD) and the mean square error (MSE) similar to the work in [55], represented as

$$\begin{cases} L_{\mathrm{SAD}} = \dfrac{1}{N}\displaystyle\sum_{i=1}^{N} \arccos\left(\dfrac{\langle \boldsymbol{Y}_i, \hat{\boldsymbol{Y}}_i\rangle}{\|\boldsymbol{Y}_i\|_2 \|\hat{\boldsymbol{Y}}_i\|_2}\right) \\ L_{\mathrm{MSE}} = \dfrac{1}{N}\displaystyle\sum_{i=1}^{N} \|\boldsymbol{Y}_i - \hat{\boldsymbol{Y}}_i\|_2^2 \end{cases} \tag{22}$$

where $\boldsymbol{Y}_i$ is the ith observed HSI pixel, and $\hat{\boldsymbol{Y}}_i$ is the ith reconstructed HSI pixel in $\hat{\boldsymbol{Y}} = \boldsymbol{MS}$. Moreover, since most mixed pixels only contain several materials (a small number of endmembers), the obtained abundance matrix is usually sparse. To this end, we introduce the ℓ_q-norm regularization term to control the sparsity of the abundance matrix ($0 < q \leq 1$) [56]. The final loss is expressed as

$$L = L_{\mathrm{SAD}} + \lambda_1 L_{\mathrm{MSE}} + \lambda_2 \|\boldsymbol{S}\|_q \tag{23}$$

where $\lambda_1 > 0$ and $\lambda_2 > 0$ are hyperparameters. Overall, Algorithm 1 provides a complete pseudocode of the proposed MSGA-CD.

Algorithm 1 Proposed MSGA-CD

Input: $\boldsymbol{Y}$, $\boldsymbol{X}_\tau$, $\boldsymbol{A}_\tau$, maximal iterations T.
Output: Endmembers $\boldsymbol{M}$, Abundances $\boldsymbol{S}$.

```
Initialize set t = 1 and randomly initialize network weight parameters;
while t ≤ T do
    Compute the global abundance matrix S_G by Eqs. (12)-(15);
    Calculate the local abundance matrix S_C by Eq. (16);
    Obtain the abundance matrix S by Eqs. (17)-(19);
    Acquire the endmember matrix M by Eqs. (20)-(21);
    Calculate the loss L by Eq. (23);
    Update weight parameters via the Adam algorithm [57];
    t ← t + 1;
end
```

IV. Experiments

A. Datasets

Four HSI unmixing datasets, one simulated dataset (called "Synthetic")[1] and three real datasets,[2] are used to verify the MSGA-CD framework.

1) Synthetic: This dataset consists of 60 × 60 pixels and 200 channels. It has five endmember classes: asphalt, concrete, limestone, conifer, and basalt. In addition, this dataset provides ground-truth data, including endmembers and abundances, to quantitatively evaluate the accuracy of unmixing.

2) Samson: This dataset has 952×952 pixels and 156 channels. This article focuses on a region of 95×95 pixels, which is identical to [39]. It includes three distinct endmembers: soil, tree, and water. Moreover, this dataset does not have ground truths but contains ground references used to assess the accuracy of unmixing quantitatively.

3) Jasper Ridge: This dataset has 512 × 614 pixels and 224 channels. This article uses a subregion of 100×100 pixels and retains 198 bands, excluding channels $1-3$, $108-112$, $154-166$, and $220-224$ due to dense water vapor and atmospheric influence. It identifies four main endmembers: tree, water, soil, and road. This dataset also includes ground references.

4) Urban: This dataset consists of 307 × 307 pixels and 210 bands. To ensure data quality, we chose 162 bands for experiments after removing noisy and corrupted bands. It has five different endmembers: asphalt, grass, tree, roof, and dirt. Furthermore, this dataset also provides ground references to evaluate unmixing performance.

B. Experimental Setting

1) Implementation Details: All experiments are run on a Microsoft Windows server with an NVIDIA GeForce GTX 1080 GPU. The experimental environment is Python version 3.7.16 and PyTorch version 1.9.1. In the experiments, we set the scale hyperparameter $\Gamma = 3$ and the superpixel

[1] http://speclib.jpl.nasa.gov/
[2] https://rslab.ut.ac.ir/data

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TABLE I
NETWORK CONFIGURATION IN THE EXPERIMENTS

Parameter	Synthetic	Samson	Jasper Ridge	Urban
Learning rate	0.0002	0.0001	0.0001	0.0003
Epochs	500	300	300	300
The number of neurons in GAT of MSGAA	16	16	16	32
k	8	8	8	8
Size of filters in CNN	3×3	3×3	3×3	3×3
The number of neurons in ACDE	64	64	16	64
λ_1	0.1	0.1	0.1	0.1
λ_2	0.002	0.002	0.002	0.002
q	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
Γ	3	3	3	3

number $N_1 = 2N_2 = 4N_3$ empirically. For the synthetic dataset, set $N_3 = 100$. For the Samson, Jasper Ridge, and Urban datasets, set $N_3 = 400, 800, 2000$, respectively. The detailed network configuration of MSGA-CD is listed in Table I. The Adam optimizer [57] is used during training, and all weights are randomly initialized.

2) Evaluation Indicators: To assess the effectiveness of MSGA-CD in HU, we employ two metrics: endmember SAD and abundance root MSE (RMSE) [32]. The former quantifies the angular difference between endmembers. The latter assesses the discrepancy between the estimated and ground truth or ground reference abundance maps. The smaller values of these indicators illustrate better performance in HU. These two indicators are described as

$$\begin{cases} \text{SAD} = \dfrac{1}{P}\displaystyle\sum_{i=1}^{P} \arccos\left(\dfrac{\langle \hat{\boldsymbol{M}}_i, \boldsymbol{M}_i \rangle}{\|\hat{\boldsymbol{M}}_i\|_2 \|\boldsymbol{M}_i\|_2}\right) \\ \text{RMSE} = \dfrac{1}{N}\displaystyle\sum_{j=1}^{N} \sqrt{\|\hat{\boldsymbol{S}}_j - \boldsymbol{S}_j\|_2^2} \end{cases} \tag{24}$$

where $\boldsymbol{M}_i$ is the ith endmember extracted by MSGA-CD, while $\hat{\boldsymbol{M}}_i$ refers to the corresponding reference endmember. Similarly, $\boldsymbol{S}_j$ is the jth abundance vector estimated by MSGA-CD, and $\hat{\boldsymbol{S}}_j$ corresponds to the reference abundance. All results (MSADs and MRMSEs) are determined by taking the average of ten runs, where the bold represents the best result and the underline is the suboptimal result.

3) Comparison Methods: To test the effectiveness of MSGA-CD, we compare some different existing HU methods in deep learning, including convolutional AE unmixing network (CNNAEU) [32], CyCU-Net [33], MSNet [34], Transformer unmixing network (DeepTrans) [36], graph attention convolutional autoencoder unmixing network (GACAE) [39], spatial–spectral attention bilateral unmixing network (SSABN) [58], U-shaped Transformer unmixing network (UST-Net) [59], AE based on multilinear mixture model (MLM-AE) [60], and DAAN [43]. All methods use default hyperparameters from the original works.

C. Unmixing Performance Comparison

1) Comparison on Synthetic: We conduct four experiments on the simulated dataset: one case without noise (signal noise ratio SNR = 0 dB) and three cases with added noise levels of SNR = 10, 20, and 30 dB to evaluate the impact of noise. The results, as shown in Table II, demonstrate that MSGA-CD consistently achieves optimal performance across all scenarios. Even as noise levels increase, MSGA-CD's overall performance remains stable with minor variations. This indicates that the proposed model is robust against noise, effectively maintaining high accuracy in endmember extraction and abundance mapping. Visualizations of endmembers and abundances for the SNR = 20 dB case are presented in Figs. 5 and 6, respectively, illustrating the model's superior performance under noisy conditions.

2) Comparison on Samson: Table III lists the quantitative results of the Samson dataset. As can be seen, MSGA-CD achieves superior performance in overall endmember extraction (see SAD) compared to other methods, though it performs suboptimal results in soil endmember extraction (slightly behind UST-Net). This reveals that the proposed ACDE module effectively improves the accuracy of endmember extraction by incorporating abundance information and original HSI information as priors into the learning process. Additionally, according to RMSEs on abundance, we observe that MSGA-CD outperforms other methods, demonstrating that MSGAA extracts a more diverse set of features and LGAF effectively integrates local and global abundance information, enhancing the accuracy of abundance extraction. Figs. 7 and 8 show the visualizations of extracted abundance maps and endmembers, respectively. It can be seen that those comparison methods relying on simple weight learning and linear constraints of pixel points result in significant deviations from reference endmember values. In contrast, the proposed ACDE module effectively improves the accuracy and stability of endmember extraction by integrating prior information. In particular, when extracting abundance maps, our method produces results closer to the reference maps for soil and trees, when compared with other methods. However, the abundance fraction for water shows some differences. This may be because water occupies a large area in the entire HSI, and changes in environmental conditions and climate, such as temperature and surface characteristics, lead to variations in spectral information. The same substance may exhibit different spectral information due to the spectral variability [61]. While MSGA-CD may struggle to distinguish subtle differences caused by spectral variability, it achieves optimal results in terms of the MRMSE abundance fractions for all substances.

3) Comparison on Jasper Ridge: Compared to Samson, this dataset has greater spatial complexity and contextual information. Even though the same materials may be located far apart in the HSI, MSGA-CD still demonstrates excellent endmember and abundance results, as shown in Table IV. This largely benefits from the proposed ACDE and MSGAA. Further, visualizations of endmembers and abundances are available in Figs. 9 and 10, respectively. It can be seen that MSGA-CD generates better endmembers. From the abundance maps, the tree and soil on either side of the water are significantly far away. CNN-based methods struggle to capture such long-range dependencies on spatial positions. In contrast, our method through MSGAA effectively extracts this information. Moreover, the abundance of water is still affected by spectral variability, and our method performs

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TABLE II
QUANTITATIVE COMPARISON ON SYNTHETIC

SNR	Metric	CNNAEU [32]	CyCU-Net [33]	MSNet [34]	DeepTrans [36]	GACAE [39]	SSABN [58]	UST-Net [59]	MLM-AE [60]	DAAN [43]	MSGA-CD(Ours)
0 dB	MSAD	0.0340±0.0066	0.0676±0.0004	0.0345±0.0159	0.0737±0.0343	0.0466±0.0014	0.0226±0.0022	0.0209±0.0017	0.0470±0.0166	0.0205±0.0021	**0.0178±0.0005**
	MRMSE	0.1540±0.0305	0.0641±0.0007	0.1085±0.0163	0.1221±0.0267	0.0761±0.0035	0.0780±0.0032	0.0754±0.0030	0.1073±0.0230	0.0571±0.0061	**0.0526±0.0008**
10 dB	MSAD	0.0387±0.0032	0.0734±0.0030	0.0390±0.0107	0.1081±0.0272	0.0717±0.0044	0.0287±0.0023	0.0286±0.0034	0.0829±0.0368	0.0258±0.0041	**0.0243±0.0007**
	MRMSE	0.1766±0.0176	0.0692±0.0008	0.1215±0.0111	0.1674±0.0311	0.0851±0.0051	0.0834±0.0049	0.0827±0.0052	0.1343±0.0214	0.0587±0.0010	**0.0564±0.0012**
20 dB	MSAD	0.0383±0.0087	0.0694±0.0009	0.0378±0.0108	0.1051±0.0410	0.0605±0.0025	0.0240±0.0024	0.0240±0.0042	0.0701±0.0335	0.0198±0.0003	**0.0188±0.0008**
	MRMSE	0.1669±0.0286	0.0659±0.0014	0.1117±0.0192	0.1439±0.0320	0.0803±0.0043	0.0813±0.0042	0.0815±0.0049	0.1243±0.0223	0.0550±0.0014	**0.0549±0.0010**
30 dB	MSAD	0.0349±0.0073	0.0685±0.0011	0.0368±0.0125	0.0762±0.0256	0.0523±0.0021	0.0233±0.0023	0.0242±0.0036	0.0498±0.0192	0.0198±0.0011	**0.0185±0.0012**
	MRMSE	0.1601±0.0187	0.0647±0.0011	0.1095±0.0187	0.1244±0.0250	0.0771±0.0063	0.0787±0.0031	0.0780±0.0042	0.1136±0.0251	0.0543±0.0031	**0.0537±0.0013**

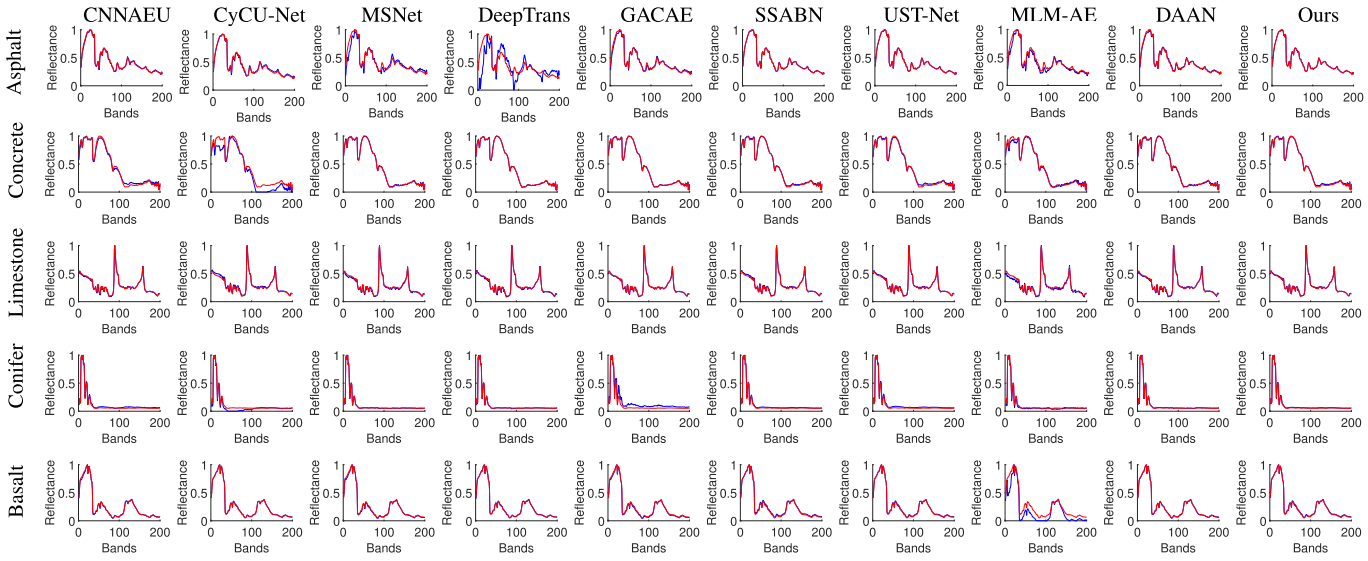


Fig. 5. Comparison of extracted endmembers on Synthetic with SNR = 20 dB, with the blue line representing the results of various methods and the red line being ground truth.

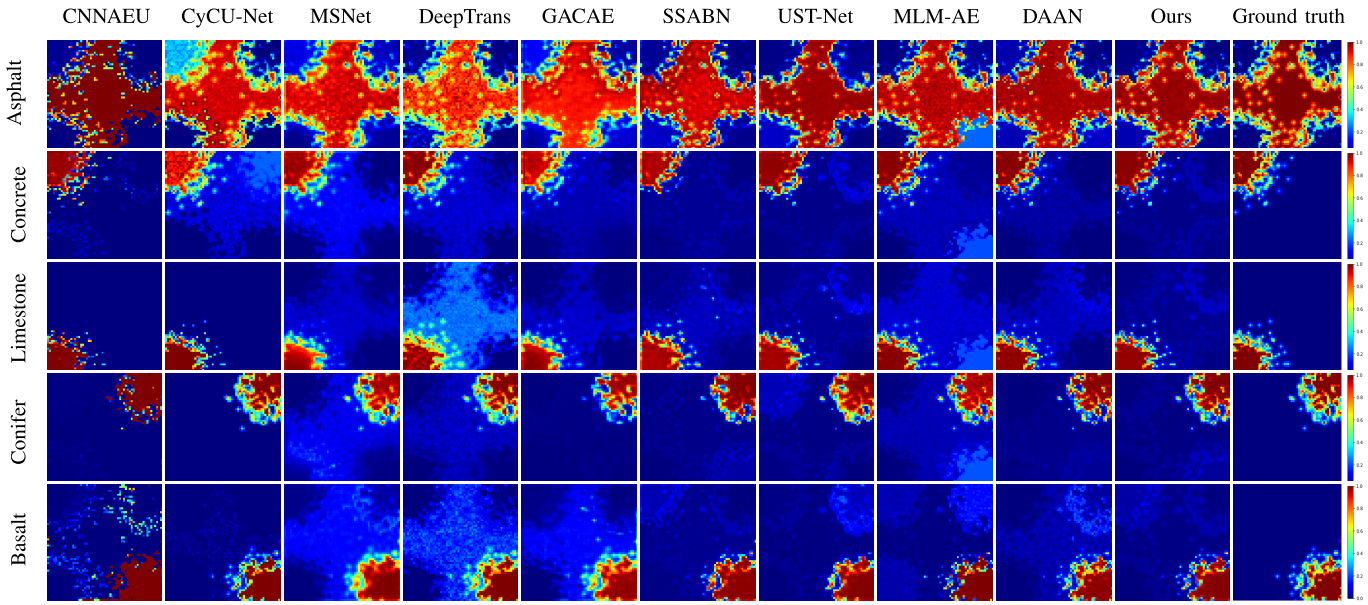


Fig. 6. Visualizations of abundance maps on Synthetic with SNR = 20 dB.

moderately in extracting the abundance of water. Nonetheless, MSGA-CD achieves a significant improvement in MRMSE of abundance.

4) Comparison on Urban: This dataset is larger than the previous two, with more complex land-object relationships and a greater variety of mixed materials. The same material is distributed in distant locations across the entire HSI rather than confined to local regions. As shown in Table V, MSGA-CD remains highly competitive. We achieve optimal results in the overall MSAD for endmembers and the MRMSE for abundance maps. Also, most materials have optimal or suboptimal results in terms of endmembers and abundance, except

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TABLE III

QUANTITATIVE COMPARISON ON SAMSON

Method	CNNAEU [32]	CyCU-Net [33]	MSNet [34]	DeepTrans [36]	GACAE [39]	SSABN [58]	UST-Net [59]	MLM-AE [60]	DAAN [43]	MSGA-CD(Ours)
					SAD					
Soil	0.0245±0.0069	0.0346±0.0086	0.0225±0.0061	0.0402±0.0354	0.0405±0.0001	0.0301±0.0260	0.0129±0.0049	0.0214±0.0062	**0.0108±0.0004**	0.0175±0.0032
Tree	0.0403±0.0022	0.1117±0.0240	0.0406±0.0048	0.0432±0.0085	0.0713±0.0001	0.0407±0.0061	0.0338±0.0013	0.0508±0.0074	0.0518±0.0013	**0.0286±0.0026**
Water	0.0725±0.0130	0.1342±0.0351	0.0642±0.0124	0.0405±0.0097	0.1308±0.0006	0.0427±0.0154	0.0356±0.0071	0.0431±0.0059	0.0356±0.0052	**0.0261±0.0031**
MSAD	0.0458±0.0058	0.0935±0.0084	0.0424±0.0058	0.0413±0.0116	0.0809±0.0002	0.0378±0.0059	0.0274±0.0023	0.0384±0.0044	0.0327±0.0015	**0.0241±0.0012**
					RMSE					
Soil	0.2555±0.0331	0.2048±0.0113	0.1133±0.0229	0.1520±0.0111	0.2182±0.0245	0.1027±0.0421	0.0809±0.0100	0.1230±0.0307	0.1264±0.0015	**0.0755±0.0073**
Tree	0.2119±0.0253	0.1803±0.0070	0.0707±0.0259	0.1387±0.0109	0.1918±0.0187	0.0948±0.0419	0.0774±0.0073	0.0718±0.0081	**0.0656±0.0020**	0.0687±0.0067
Water	0.2307±0.0323	0.2160±0.0223	0.0676±0.0183	0.2315±0.0117	0.2159±0.0480	**0.0344±0.0063**	0.0793±0.0143	0.0828±0.0255	0.1269±0.0011	0.0715±0.0114
MRMSE	0.2327±0.0170	0.2004±0.0032	0.0839±0.0137	0.1741±0.0056	0.2086±0.0266	0.0773±0.0275	0.0791±0.0092	0.0925±0.0199	0.1063±0.0011	**0.0719±0.0052**

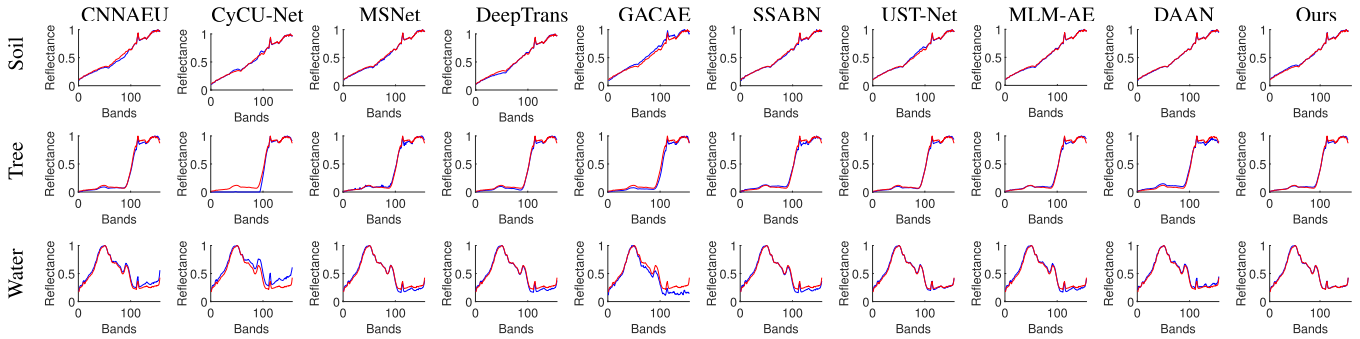


Fig. 7. Comparison of extracted endmembers on Samson, with the blue line representing the results of various methods and the red line being the ground reference.

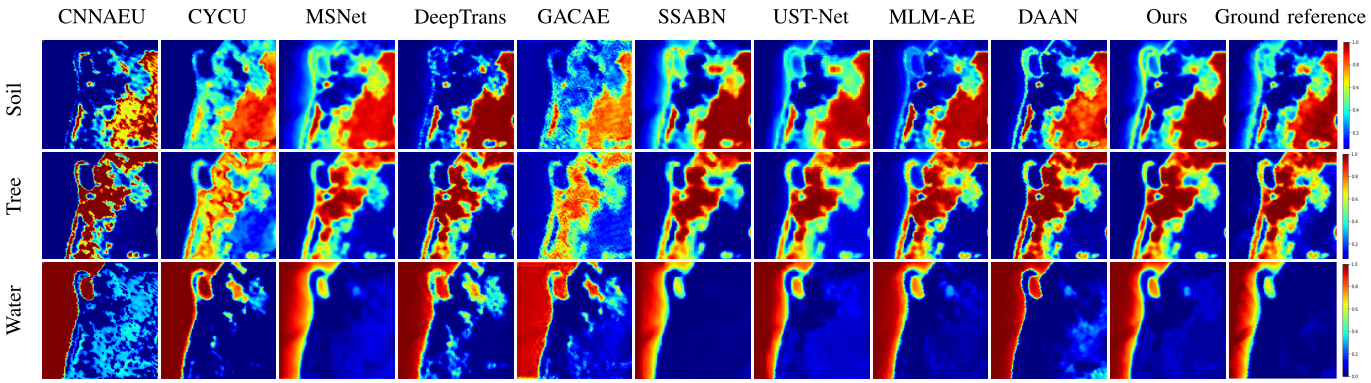


Fig. 8. Visualizations of abundance maps on Samson.

TABLE IV

QUANTITATIVE COMPARISON ON JASPER RIDGE

Method	CNNAEU [32]	CyCU-Net [33]	MSNet [34]	DeepTrans [36]	GACAE [39]	SSABN [58]	UST-Net [59]	MLM-AE [60]	DAAN [43]	MSGA-CD(Ours)
					SAD					
Tree	0.1758±0.0103	0.0924±0.0343	0.1452±0.0320	0.0597±0.0244	0.1486±0.0003	0.0515±0.0173	0.0649±0.0111	0.1789±0.0375	0.1477±0.0002	**0.0272±0.0055**
Water	0.0521±0.0036	0.1256±0.0015	0.1215±0.0219	0.0787±0.0347	0.1028±0.0032	0.0480±0.0128	**0.0353±0.0073**	0.1002±0.0102	0.0699±0.0053	0.0363±0.0041
Soil	0.1335±0.0192	0.0618±0.0086	0.1923±0.0433	0.0698±0.0224	0.1167±0.0003	0.0904±0.0503	0.0809±0.0381	0.0811±0.0110	0.1192±0.0009	**0.0341±0.0113**
Road	0.1010±0.0114	0.1728±0.0115	0.1404±0.0403	0.0840±0.0420	0.0900±0.0002	0.0961±0.0340	0.0883±0.0483	0.1194±0.0209	0.0883±0.0005	**0.0239±0.0023**
MSAD	0.0906±0.0057	0.1132±0.0093	0.1499±0.0116	0.0730±0.0097	0.1145±0.0008	0.0715±0.0167	0.0674±0.0160	0.1199±0.0089	0.1062±0.0014	**0.0304±0.0026**
					RMSE					
Tree	0.2635±0.0322	0.1086±0.0115	0.1084±0.0346	0.0803±0.0194	0.1465±0.0186	0.1599±0.0667	0.1001±0.0303	0.2485±0.0144	0.0951±0.0019	**0.0647±0.0047**
Water	0.2091±0.0270	0.0622±0.0036	0.0960±0.0373	0.0781±0.0122	0.1199±0.0337	0.0483±0.0147	**0.0378±0.0049**	0.1328±0.0423	0.1357±0.0037	0.0881±0.0116
Soil	0.3586±0.0395	0.2343±0.0081	0.1713±0.0200	0.1700±0.0247	0.1752±0.0176	0.2033±0.0647	0.1350±0.0382	0.1697±0.0393	0.1434±0.0038	**0.1129±0.0085**
Road	0.2851±0.0166	0.2335±0.0014	0.1396±0.0261	0.1586±0.0294	0.1602±0.0256	0.1638±0.0404	0.1289±0.0457	0.1935±0.0245	0.1324±0.0006	**0.0900±0.0061**
MRMSE	0.2791±0.0228	0.1597±0.0050	0.1288±0.0126	0.1218±0.0118	0.1504±0.0061	0.1438±0.0425	0.1004±0.0266	0.1861±0.0165	0.1266±0.0011	**0.0889±0.0017**

for the roof material performing moderately. This is likely due to the variability in roof spectra caused by factors such as illumination angle, weather changes, and material aging, which increase spectral variability and introduce bias into endmember learning. Moreover, the visualizations of endmembers and abundance maps are shown in Figs. 11 and 12, respectively. The abundance maps show that each material is broadly dispersed across the HSI. CNN-based methods struggle to capture global abundance information, while single-scale GNNs like GACAE perform poorly. MSGA-CD extracts global

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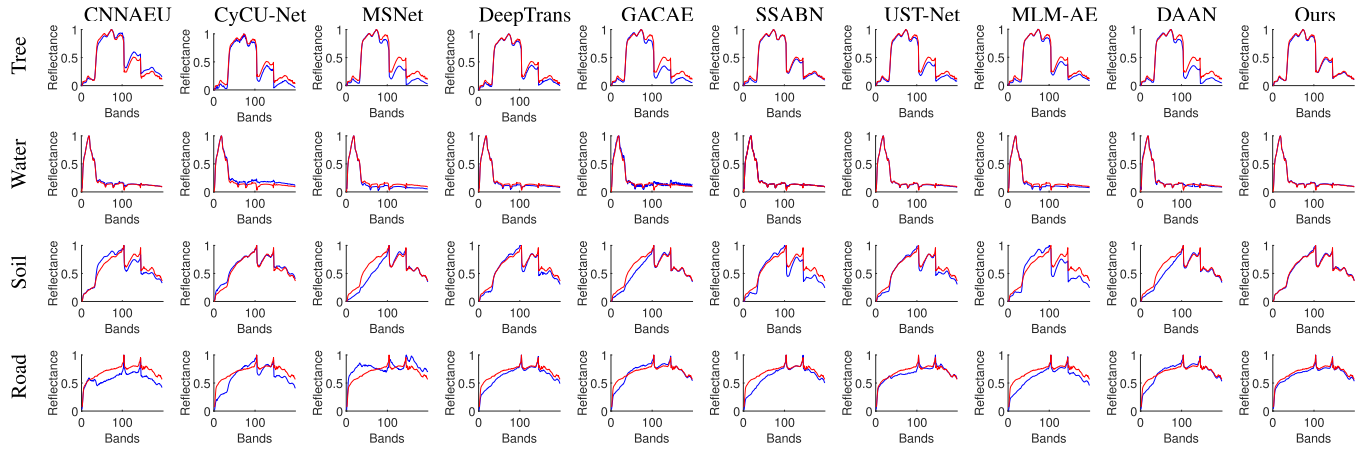


Fig. 9. Comparison of extracted endmembers on Jasper Ridge, with the blue line representing the results of various methods and the red line being the ground reference.

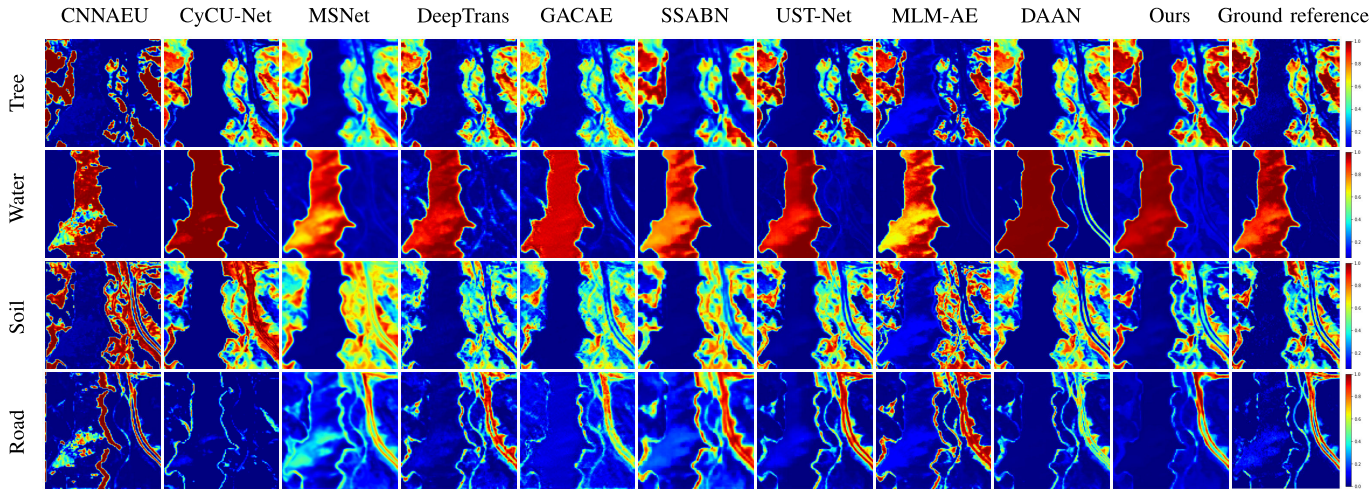


Fig. 10. Visualizations of abundance maps on Jasper Ridge.

TABLE V
QUANTITATIVE COMPARISON ON URBAN

Method	CNNAEU [32]	CyCU-Net [33]	MSNet [34]	DeepTrans [36]	GACAE [39]	SSABN [58]	UST-Net [59]	MLM-AE [60]	DAAN [43]	MSGA-CD(Ours)
					SAD					
Asphalt	**0.0363±0.0024**	0.2196±0.1553	0.1920±0.1123	0.0836±0.0170	0.2483±0.0005	0.0499±0.0105	0.0495±0.0072	0.1281±0.0320	0.0735±0.0450	0.0425±0.0060
Grass	0.0633±0.0056	0.1255±0.0456	0.1754±0.0976	0.1549±0.0631	0.3280±0.0004	0.0658±0.0349	0.0522±0.0148	0.0447±0.0044	0.0618±0.0348	**0.0412±0.0021**
Tree	0.1159±0.0227	0.1322±0.0267	0.2668±0.0216	0.1474±0.0150	0.2147±0.0002	0.1186±0.0197	0.1081±0.0145	0.2064±0.0068	0.1494±0.0361	**0.0793±0.0215**
Roof	0.0402±0.0046	0.1047±0.0130	0.2028±0.1793	0.1029±0.0286	**0.0354±0.0001**	0.0716±0.0435	0.1436±0.1238	0.4337±0.0100	0.2014±0.1554	0.1140±0.0105
Dirt	0.1066±0.0122	**0.0206±0.0011**	0.1630±0.0501	0.0886±0.0119	0.1525±0.0010	0.1855±0.0260	0.2118±0.0371	0.2154±0.0071	0.1993±0.0457	0.0736±0.0229
MSAD	0.0724±0.0057	0.1205±0.0153	0.2000±0.0412	0.1155±0.0053	0.1958±0.0001	0.0983±0.0145	0.1130±0.0238	0.2057±0.0062	0.1371±0.0477	**0.0701±0.0035**
					RMSE					
Asphalt	0.2464±0.0198	0.2969±0.0128	0.1796±0.0254	0.2796±0.0179	0.2149±0.0002	0.1926±0.0302	0.1787±0.0412	**0.1399±0.0094**	0.1909±0.0444	0.1427±0.0064
Grass	0.3053±0.0166	0.3291±0.0072	0.2264±0.0205	0.2735±0.0141	0.2767±0.0004	0.1829±0.0262	0.1840±0.0433	0.2161±0.0153	0.1977±0.0320	**0.1745±0.0162**
Tree	0.2022±0.0126	0.2813±0.0338	0.1873±0.0121	0.1974±0.0194	0.2026±0.0004	0.1433±0.0265	**0.0944±0.0116**	0.1353±0.0199	0.1422±0.0236	0.1053±0.0024
Roof	0.1424±0.0122	0.1825±0.0031	0.1076±0.0077	0.1663±0.0022	0.1722±0.0004	0.0898±0.0095	**0.0707±0.0165**	0.0910±0.0161	0.0879±0.0107	0.0833±0.0013
Dirt	0.2613±0.0345	0.2187±0.0046	0.1744±0.0225	**0.1516±0.0141**	0.2016±0.0004	0.1650±0.0280	0.1920±0.0397	0.1739±0.0130	0.1802±0.0309	0.1554±0.0092
MRMSE	0.2315±0.0096	0.2617±0.0056	0.1751±0.0104	0.2137±0.0066	0.2136±0.0001	0.1547±0.0175	0.1440±0.0225	0.1512±0.0101	0.1598±0.0181	**0.1322±0.0038**

abundance information from MSGAA and combines it with local abundance information extracted by CNN. This information is effectively fused using LGAF, resulting in a more comprehensive abundance representation. Therefore, the difference between MSGA-CD and the reference abundance maps is minor. Besides, the endmember maps reveal significant differences between those methods based on simple weight learning and the reference endmember values. In contrast, MSGA-CD using ACDE achieves superior results.

D. Ablation Studies

To verify the effectiveness of our proposed method, we conducted a series of ablation experiments. As shown in Table VI, we compare the roles of three modules: MSGAA, LGAF, and ACDE. In the ablation studies of LGAF, we employ a direct addition fusion strategy. For the ACDE module, a 1×1 convolution replaces it. Additionally, we investigate the impact of CNN and single-scale graph attentional abundance

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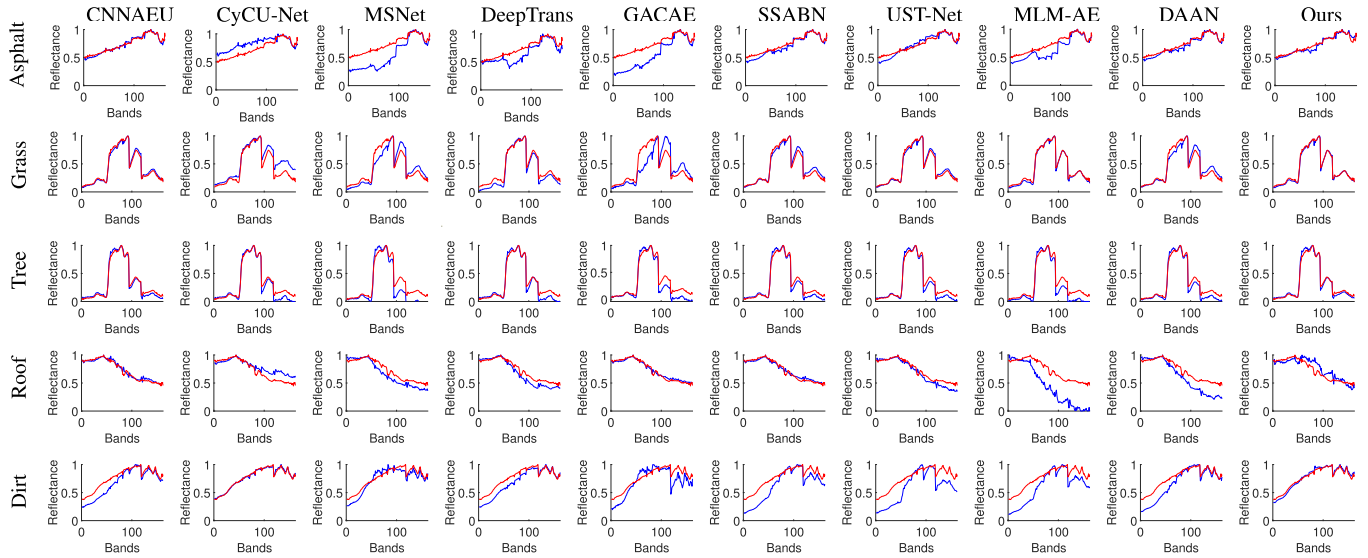


Fig. 11. Comparison of extracted endmembers on Urban, with the blue line representing the results of various methods and the red line being ground reference.

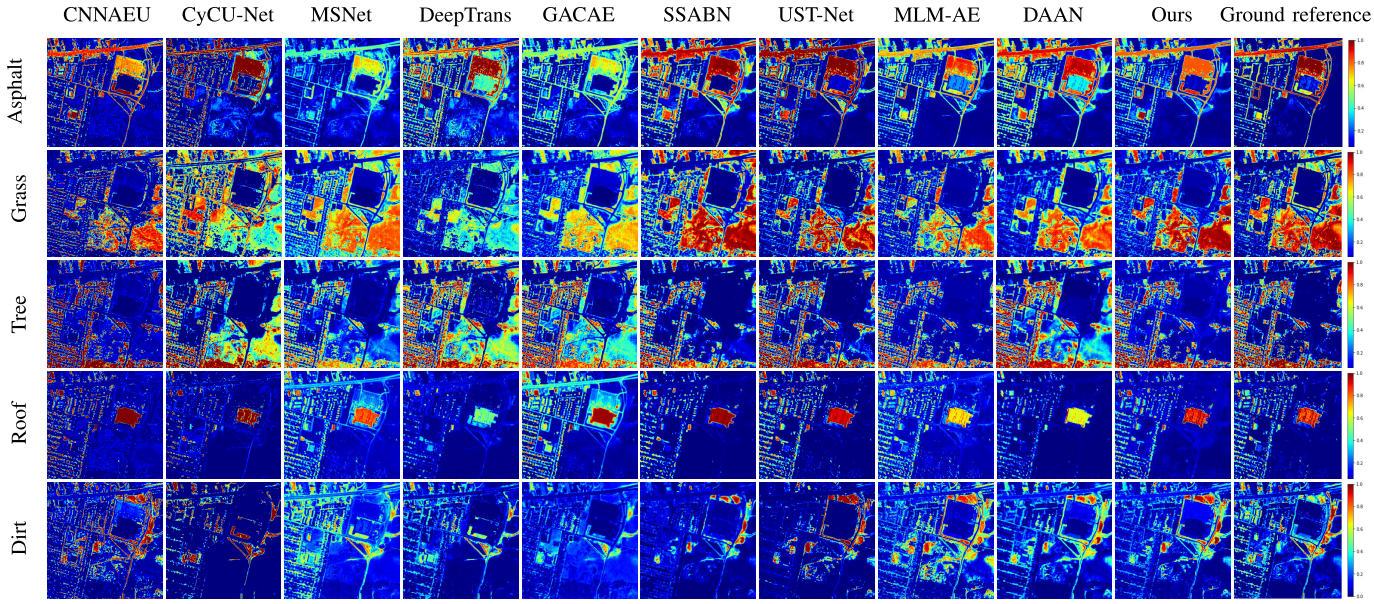


Fig. 12. Visualizations of abundance maps on Urban.

TABLE VI
RESULTS OF ABLATION STUDIES

	Module					Samson		Jasper Ridge		Urban	
Model	CNN	SSGAA	MSGAA	LGAF	ACDE	MSAD	MRMSE	MSAD	MRMSE	MSAD	MRMSE
1	✓				✓	0.0308±0.0061	0.1088±0.0130	0.0554±0.0060	0.1142±0.0070	0.0895±0.0009	0.1395±0.0005
2			✓		✓	0.0311±0.0011	0.1070±0.0006	0.0375±0.0037	0.1378±0.0011	0.0707±0.0030	0.1652±0.0009
3	✓		✓	✓		0.0957±0.0037	0.1194±0.0111	0.1098±0.0002	0.1140±0.0010	0.1172±0.0021	0.1626±0.0007
4	✓		✓		✓	0.0255±0.0046	0.1065±0.0015	0.0529±0.0023	0.1781±0.0005	0.0806±0.0009	0.1473±0.0023
5	✓	✓		✓	✓	0.0305±0.0053	0.0776±0.0040	0.0372±0.0047	0.0998±0.0044	0.0748±0.0019	0.1347±0.0045
MSGA-CD (Ours)	✓		✓	✓	✓	**0.0241±0.0012**	**0.0719±0.0052**	**0.0304±0.0026**	**0.0889±0.0017**	**0.0701±0.0035**	**0.1322±0.0038**

(SSGAA) with $\Gamma = 1$. We conduct experiments on three real datasets while keeping other hyperparameters constant.

From the experimental results of Model 1, Model 2, and MSGA-CD, it is evident that combining the global abundance information extracted by MSGAA with the local abundance information captured by CNN, after processing through LGAF, significantly improves the accuracy of the abundance information. This demonstrates the complementary advantages of the local spatial features captured by CNN and the long-range dependency on spatial positions captured by GNN. The combination of these two types of information greatly enhances the accuracy of abundance estimation. Meanwhile,

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TABLE VII
INFLUENCE OF Γ

	Samson		Jasper Ridge		Urban	
Γ	MSAD	MRMSE	MSAD	MRMSE	MSAD	MRMSE
1	0.0305±0.0053	0.0776±0.0040	0.0372±0.0047	0.0998±0.0044	0.0748±0.0019	0.1347±0.0045
2	0.0296±0.0035	0.0737±0.0032	0.0363±0.0028	0.0970±0.0052	0.0743±0.0022	0.1344±0.0026
3	**0.0241±0.0012**	**0.0719±0.0052**	**0.0304±0.0026**	**0.0889±0.0017**	**0.0701±0.0035**	**0.1321±0.0038**
4	0.0272±0.0024	0.0733±0.0030	0.0332±0.0015	0.0923±0.0013	0.0721±0.0017	0.1340±0.0025
5	0.0263±0.0033	0.0755±0.0035	0.0358±0.0037	0.0985±0.0031	0.0752±0.0013	0.1334±0.0028

with the introduction of prior information into endmember learning through ACDE, the overall endmember performance is also extremely enhanced, as evidenced by the comparison of Model 3 and MSGA-CD. This confirms that ACDE's use of spectral prior information helps refine endmember estimation, improving the accuracy and reliability of endmember extraction. Further, from Model 4 and MSGA-CD, we can see that LGAF affects the performance of abundance and endmembers compared with the fusion strategy of direct addition. This illustrates the importance of a dedicated fusion mechanism that effectively combines the strengths of different information. Moreover, when MSGA-CD degenerates to the single-scale case (Model 5), the MSADs on endmembers reduce slightly and the MRMSEs on abundance decrease greatly. This reveals the importance of a multiscale strategy. This reveals the importance of a multiscale strategy in capturing diverse spatial dependencies for more robust unmixing results. Overall, these analyses validate the effectiveness of the proposed modules.

E. Hyperparameter Discussion

This section discusses hyperparameters of MSGA-CD, the number of scales Γ and the generated superpixel N_τ. Since it is challenging to determine these hyperparameters adaptively, we conduct extensive experiments and discussions within a certain range based on empirical experience.

1) **The Influence of Γ:** We consider different Γ values ranging from 1 to 5 on three real datasets (Samson, Jasper Ridge, and Urban). The results are listed in Table VII. It can be seen that the unmixing performance initially improves as the number of scales increases, reaching the best results with $\Gamma = 3$. Beyond this scale, performance declines, which may introduce redundancy, affecting the model's generalization capability. When Γ is small, the feature diversity is limited. Therefore, an appropriate Γ is required to enhance model performance. Based on these analyses, we empirically set $\Gamma = 3$ for all three real datasets.
2) **The Impact of N_3:** We will discuss the number of N_3 in detail. Specifically, we empirically set $N_1 = 2N_2 = 4N_3$ for simplicity. As shown in Table VIII, we run 10 times on Samson from $N_3 = 100$ to $N_3 = 1000$. The results show that when $N_3 = 400$, $N_2 = 800$, and $N_1 = 1600$, our network produces the best overall abundance maps and endmember extraction results. Similarly, as listed in the results on the Jasper Ridge data with N_3 ranging from 100 to 1000, MSGA-CD performs best in the overall abundance maps and endmember extraction results when $N_3 = 800$, $N_2 = 1600$, and $N_1 = 3200$. For the Urban dataset, we also analyze N_3 from 500 to 5000. When $N_3 = 2000$ and $N_3 = 5000$, we obtain the optimal endmember and abundance results, respectively. However, the MRMSE with $N_3 = 2000$ is slightly higher than $N_3 = 5000$. We ultimately determine $N_3 = 2000$, $N_2 = 4000$, and $N_1 = 8000$ empirically.

TABLE VIII
IMPACT OF N_3

	Samson			Jasper Ridge			Urban	
N_3	MSAD	MRMSE	N_3	MSAD	MRMSE	N_3	MSAD	MRMSE
100	0.0312±0.0034	0.0779±0.0033	100	0.0479±0.0050	0.1061±0.0028	500	0.0720±0.0013	0.1392±0.0042
200	0.0289±0.0029	0.0773±0.0035	200	0.0439±0.0029	0.1044±0.0022	1000	0.0736±0.0013	0.1367±0.0042
300	0.0271±0.0019	0.0748±0.0032	300	0.0378±0.0033	0.1038±0.0031	1500	0.0740±0.0020	0.1346±0.0030
400	**0.0241±0.0012**	**0.0719±0.0052**	400	0.0398±0.0032	0.0979±0.0028	2000	**0.0701±0.0035**	0.1322±0.0038
500	0.0261±0.0024	0.0735±0.0037	500	0.0370±0.0030	0.1004±0.0037	2500	0.0734±0.0012	0.1330±0.0031
600	0.0248±0.0010	0.0731±0.0045	600	0.0358±0.0037	0.0991±0.0035	3000	0.0779±0.0014	0.1320±0.0047
700	0.0272±0.0031	0.0746±0.0031	700	0.0326±0.0018	0.0926±0.0013	3500	0.0785±0.0017	0.1321±0.0038
800	0.0262±0.0048	0.0751±0.0028	800	**0.0304±0.0026**	**0.0889±0.0017**	4000	0.0787±0.0013	0.1328±0.0046
900	0.0274±0.0026	0.0755±0.0032	900	0.0309±0.0026	0.0934±0.0006	4500	0.0788±0.0014	0.1331±0.0043
1000	0.0261±0.0010	0.0753±0.0029	1000	0.0368±0.0024	0.0972±0.0036	5000	0.0773±0.0012	**0.1320±0.0045**

V. CONCLUSION

This article proposes an MSGA-CD for HU, which consists of three key components: MSGAA, LGAF, and ACDE. The MSGAA module integrates a multiscale strategy into GAT to obtain global long-range dependency information on spatial positions at different levels, thereby enhancing the diversity of features. Then, the LGAF module fuses the local abundance information obtained by CNN with the global abundance information by MSGAA to learn a more accurate abundance representation. Finally, ACDE uses the abundance fraction to cluster the pixels of the HSI to drive endmember learning, effectively improving the accuracy of endmember extraction. Experiments validate that MSGA-CD outperforms other advanced methods and provides a promising method for HU.

However, the proposed method is limited by spectral variability, i.e., the spectral characteristics of the same material may vary due to factors such as environmental changes, leading to misidentification and deviations in abundance estimation. In future work, we will plan to design an effective loss to improve the impact of spectral variability on HSI unmixing performance.

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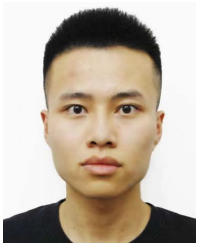
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